Mathematical Statistics II

Point Estimation: Principles and Theory

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1 Introduction

Introduction

- We now introduce principles of parameter estimation, as well as associated theory.
- The three approaches we have considered thus far (MoM, MLE, Bayes) are great starting points, but we would like to study general properties of parameter estimations, and how these approaches align with these properties.
- There is no "correct" order in which these material are presented, so we deviate a bit from our textbooks.
- This work is largely based on Rice (2007, Sections 8.5.2, 8.7–8.8), Casella and Berger (2024, Chapters 6, 7, 10), as well as some supplemental material from Pawitan (2001).

2 Fisher's Information

Curvature of the Likelihood Function

- Both the MLE and Bayesian estimates are forms of *Likelihood based inference*.
- For the MLE, we are interested in the maximum of the likelihood function.
- For Bayesian inference, the posterior distribution is the (scaled) product of the likelihood and a prior distribution:
$$\pi(\theta|x) \propto L(\theta) \times \pi(\theta).$$

- In both instances, the shape of the likelihood function $L(\theta)$ is of interest.

  Definition: The score function

  Let $L(\theta) = f(x^*; \theta)$ be the likelihood of some model, conditioned on the observed data (x^* is assumed to be multi-variate). Then we define the score function $S(\theta)$ to be:

$$S(\theta) = \frac{\partial}{\partial \theta} \log L(\theta).$$

- Assuming the MLE $\hat{\theta}$ exists and is not a boundary point, then:

$$S\big(\hat{\theta}\big) = 0.$$

- As written, the score function is dependent on a fixed data set: x^*.
- We could also expand this by considering it as a random function of any possible data X, which is random.
- Since we are interested in the curvature of the likelihood function, we should also consider the second derivative.
- At the MLE, the second derivative of the log-likelihood will be *negative*, so it's natural to consider the negative second derivative of the log-likelihood:

$$I(\theta) = -\frac{\partial^2}{\partial \theta^2} \log L(\theta).$$

- Again, we are treating the likelihood function $L(\theta)$ as a regular (non-random) function, fixed on x^* (this is slightly different than many texts).
- The curvature at the MLE is of particular interest.

  Definition: The Observed Fisher Information

  Let $L(\theta)$ be the likelihood function, fixed on the observed data x^*. The *observed Fisher information* is the second derivative of $\log L(\theta)$, evaluated at the MLE:

$$\mathcal{I}\big(\hat{\theta}\big) := I\big(\hat{\theta}\big) = -\frac{\partial^2}{\partial \theta^2} \log L\big(\hat{\theta}\big)$$

  Note that when θ is multi-variate, this is a matrix with entries:

$$\Big[\mathcal{I}\big(\hat{\theta}\big)\Big]_{i,j} = -\frac{\partial^2}{\partial \theta_i \partial \theta_j} \log L(\theta),$$

  also denoted $\mathcal{I}\big(\hat{\theta}\big) = -\nabla^2 \log L\big(\hat{\theta}\big)$.

- Note that large values of $\mathcal{I}\big(\hat{\theta}\big)$ in the univariate case (and large eigen-values / determinant in the multivariate case) indicate a "tight peak" around the maximum, suggesting less uncertainty about the parameter θ.

  Theorem: Expected Score

  Now consider treating the score function $S(\theta)$ as a random function, dependent on random input data X. That is:

$$S(\theta) = \frac{\partial}{\partial \theta} \log f(X; \theta).$$

  Then, under mild regularity conditions, the expected score is zero:

$$E\big[S(\theta)\big] = 0.$$

Proof.

$$E\Big[\frac{\partial}{\partial\theta}\log f(X;\theta)\Big] = \int_{\mathcal{X}} \frac{\partial}{\partial\theta}\log f(x;\theta) f(x;\theta)\,dx$$

$$\begin{aligned}
&= \int_{\mathcal{X}} \frac{f(x;\theta)}{f(x;\theta)}\frac{\partial}{\partial\theta} f(x;\theta)\,dx \\
&= \int_{\mathcal{X}} \frac{\partial}{\partial\theta} f(x;\theta) \\
&= \frac{\partial}{\partial\theta}\int_{\mathcal{X}} f(x;\theta)\,dx \\
&= \frac{\partial}{\partial\theta} 1 = 0.
\end{aligned}$$

Note that the conditions needed for this is the ability to swap the integral with the derivative, which is a condition that we briefly discussed last semester (Dominated Convergence Theorem, for instance). □

Definition: Fisher's Information

Continuing with the treatment of the score function as a random function of X, we will define Fisher's Information to be the following:

$$\mathcal{I}(\theta) = E\Big[\Big(\frac{\partial}{\partial\theta}\log f(X;\theta)\Big)^2\Big] = E\Big[\big(S(\theta)\big)^2\Big].$$

Proposition: Variance of the Score

If the partial derivative and integral operations can be exchanged, then

$$\mathcal{I}(\theta) = \mathrm{Var}\big(S(\theta)\big) = -E\Big[\frac{\partial^2}{\partial\theta^2}\log f(X;\theta)\Big],$$

where the score function is treated as random.

Proof. If we can exchange integration and derivation, then the first equality in the statement is trivial. Under these conditions, we showed:

$$E\big[S(\theta)\big] = E\Big[\frac{\partial}{\partial\theta}\log f(X;\theta)\Big] = 0.$$

Therefore:

$$\begin{aligned}
\mathrm{Var}\big(S(\theta)\big) &= E\Big[S(\theta)^2\Big] - \Big(E\big[S(\theta)\big]\Big)^2 \\
&= \mathcal{I}(\theta).
\end{aligned}$$

Now if the second derivatives exist, the same argument that used to show the equality above can be used to show:

$$E\Big[\frac{\frac{\partial^2}{\partial\theta^2} f(X;\theta)}{f(X;\theta)}\Big] = \int_{\mathcal{X}} \frac{\partial^2}{\partial\theta^2} f(X;\theta)\,dx = 0.$$

(To do this, we exchange partial derivatives with the integral, note the integral is equal to one, which has partial derivative zero).

Building on these two results, note the following identity:

$$\begin{aligned}
\frac{\partial^2}{\partial\theta^2}\log f(X;\theta) &= \frac{\partial}{\partial\theta}\Big(\frac{\frac{\partial}{\partial\theta}f(X;\theta)}{f(X;\theta)}\Big) \\
&= \frac{f(X;\theta)\frac{\partial^2}{\partial\theta^2}f(X;\theta) - \big(\frac{\partial}{\partial\theta}f(X;\theta)\big)^2}{\big(f(X;\theta)\big)^2} \\
&= \frac{\frac{\partial^2}{\partial\theta^2}f(X;\theta)}{f(X;\theta)} - \Big(\frac{\frac{\partial}{\partial\theta}f(X;\theta)}{f(X;\theta)}\Big)^2 \\
&= \frac{\frac{\partial^2}{\partial\theta^2}f(X;\theta)}{f(X;\theta)} - \Big(\frac{\partial}{\partial\theta}\log f(X;\theta)\Big)^2
\end{aligned}$$

Taking expectation of both sides of this equation yields:

$$\begin{aligned}
E\Big[\frac{\partial^2}{\partial\theta^2}\log f(X;\theta)\Big] &= E\Big[\frac{\frac{\partial^2}{\partial\theta^2}f(X;\theta)}{f(X;\theta)}\Big] - E\Big[\Big(\frac{\partial}{\partial\theta}\log f(X;\theta)\Big)^2\Big] \\
&= -E\Big[\Big(\frac{\partial}{\partial\theta}\log f(X;\theta)\Big)^2\Big],
\end{aligned}$$

where the last inequality is a result of the previously derived identity

$$E\Big[\frac{\frac{\partial^2}{\partial\theta^2}f(X;\theta)}{f(X;\theta)}\Big] = 0.$$

Thus, we have proven the statement

$$\mathcal{I}(\theta) = \mathrm{Var}\big(S(\theta)\big) = -E\Big[\frac{\partial^2}{\partial\theta^2}\log f(X;\theta)\Big].$$

□

- This proposition is useful because:
  - If the log-likelihood is differentiable, the first definition $\mathcal{I}(\theta) = E[S(\theta)^2]$ always exists.
  - Typically, we can express this as $\mathcal{I}(\theta) = \mathrm{Var}(S(\theta))$, which is a nice interpretable quantity.
  - The final expression $\mathcal{I}(\theta) = -E\Big[\frac{\partial^2}{\partial\theta^2}\log f(X;\theta)\Big]$ is usually the easiest to calculate.
- For the multivariate case, this is equivalent to writing:

$$\begin{aligned}
\big[\mathcal{I}(\theta)\big]_{i,j} &:= E_\theta\Big[\frac{\partial\log f(X;\theta)}{\partial\theta_i}\frac{\partial\log f(X;\theta)}{\partial\theta_j}\Big] \\
&= \mathrm{Cov}\Big(\frac{\partial\log f(X;\theta)}{\partial\theta_i}, \frac{\partial\log f(X;\theta)}{\partial\theta_j}\Big) \\
&= -E\Big[\frac{\partial^2}{\partial\theta_i\partial\theta_j}\log f(X;\theta)\Big]
\end{aligned}$$

- Note that the curvature $I(\theta)$, defined as the second derivative of $L(\theta)$ (fixed function), exists for all points θ.
- However, we're primarily only interested in this function at the MLE, *or* when we regard $S(\theta)$ as a random function of arbitrary X (not fixed on x^*).

- Thus, it's fairly uncommon to discuss both $I(\theta)$ and $\mathcal{I}(\theta)$ (formally in), and most texts just discuss $\mathcal{I}(\theta)$.

  Note on observed fisher information

  We defined the observed fisher information as:

  $$\mathcal{I}(\hat{\theta}) := I(\hat{\theta}) = -\frac{\partial^2}{\partial\theta^2} \log L(\hat{\theta}).$$

  This matches the definition of fisher's information, because in the observed case the data $X = x^*$ is not random, and the expectation is a constant; thus, the notation is consistent.

Comment on numeric optimization
If numeric optimization is used to find the MLE, many approaches estimate the Hessian of the loss-function. If the negative log-likelihood is used, setting `optim(..., hessian = TRUE)` can be used to compute the Observed Fisher's Information (don't forget to adjust for variable transformations!)

3 Asymptotic proprties of the MLE and Bayesian estimators

Normality of MLE

- Under weak regularity conditions, both the MLE and Bayesian posterior converge to a Normal distribution, centered at the true value.
- We'll begin with a sketch proof of the distribution of the MLE.

Theorem: Asymptotic distribution of the MLE
Let $x_1^*, x_2^*, \ldots, x_n^*$ come from a family of distributions with density $f(x;\theta)$. If the true value of θ is θ_0, i.e., $X_i \overset{iid}{\sim} f(x;\theta_0)$, and we denote $\hat{\theta}$ as the MLE of θ, then

$$\sqrt{n\mathcal{I}(\theta_0)}\big(\hat{\theta} - \theta_0\big) \xrightarrow{d} Z$$

where $Z \sim N(0,1)$.

Proof. Recall in this setup, we have X being multivariate. Thus, the likelihood function is

$$L(\theta) := \prod_{i=1}^{n} f(X_i;\theta).$$

Note that here, we are considering the long-run distribution of the MLE estimator, so X_i is treated as random, not fixed.
The theorem statement depends on $\mathcal{I}(\theta_0)$, which is defined as the (expectation of the) negative second derivative of the log-likelihood, evaluated at θ_0.

$$\ell(\theta) = \sum_{i=1}^{n} \log f(X_i;\theta).$$

By definition,

$$\hat{\theta} = \operatorname*{argmax}_{\theta} \ell(\theta),$$

for any fixed $X = x^*$. Thus, if the MLE is not a boundary point, and the likelihood function is differentiable, then

$$\ell'(\hat{\theta}) = 0.$$

Now consider a Taylor-series expansion of $\ell'(\hat{\theta})$ around the point θ_0:

$$0 = \ell'(\hat{\theta}) = \ell'(\theta_0) + \big(\hat{\theta} - \theta_0\big)\ell''(\theta_0) + \frac{1}{2}\big(\hat{\theta} - \theta_0\big)^2 \ell'''(\theta^*),$$

where θ^* is some point between $\hat{\theta}$ and θ_0.
Now we want to rearrange so that on the left hand side, we get the expression that we are looking for: $\sqrt{nI(\theta_0)}\big(\hat{\theta} - \theta_0\big)$. Subtracting away terms involving $\big(\hat{\theta} - \theta_0\big)$,

$$-\big(\hat{\theta} - \theta_0\big)\Big(\ell''(\theta_0) + \frac{1}{2}\big(\hat{\theta} - \theta_0\big)\ell'''(\theta^*)\Big) = \ell'(\theta_0),$$

and multiplying by $\sqrt{n}$, we have:

$$\begin{aligned}\sqrt{n}\big(\hat{\theta} - \theta_0\big) &= \frac{\sqrt{n}\ell'(\theta_0)}{-\ell''(\theta_0) - \frac{1}{2}\big(\hat{\theta} - \theta_0\big)\ell'''(\theta^*)} \\ &= \frac{\frac{1}{\sqrt{n}}\ell'(\theta_0)}{-\frac{1}{n}\ell''(\theta_0) - \frac{1}{2n}\big(\hat{\theta} - \theta_0\big)\ell'''(\theta^*)}.\end{aligned}$$

The last step seems a bit odd, but it will be used to apply Slutsky's Theorem (covered last semester) to finish the proof. As a reminder, if W_n is a sequence of random variables that converges in *distribution* to W, and Y_n is a sequence of random variables that convergence in *probability* to a constant c, then $W_n/Y_n \xrightarrow{d} W/c$. Thus, we want something like:

$$\sqrt{n}\big(\hat{\theta} - \theta_0\big) = W_n/Y_n,$$

where W_n converges in distribution to some normally distributed random variable, and Y_n converges to a constant.
We have expressed the numerator such that:

$$\frac{1}{\sqrt{n}}\ell'(\theta_0) = \frac{1}{\sqrt{n}}\sum_{i=1}^{n}\frac{\partial}{\partial\theta}\log f(X_i;\theta_0).$$

Since $E[\ell'(\theta_0)] = E[S(\theta)] = 0$, the central limit theorem states:

$$\frac{1}{\sqrt{n}}\ell'(\theta_0) \xrightarrow{d} N(0, \mathcal{I}(\theta_0)).$$

In the denominator, the leading term converges in probability to $I(\theta_0)$ by the weak law of large numbers:

$$-\frac{1}{n}\ell''(\theta_0) \xrightarrow{p} E\Big[-\frac{\partial^2}{\partial\theta^2}\log f(X_i;\theta_0)\Big] = I(\theta_0).$$

Now we need to be a bit careful with the remainder term. The argument follows closely that which was used to prove the CLT last semester. This does involve a few "regularity" conditions, under which we can show that the remainder term converges to zero in probability. In this case, the entire denominator converges to zero in probability. The additional conditions needed are:

- bounded third derivative: $|\ell'''(\theta)| \leq M(X)$ for all θ, where $E[M(X)] < \infty$.

This last identity is needed because the remainder is:

$$R_n = \frac{1}{2n}\big(\hat{\theta} - \theta_0\big)\sum_{i=1}^{n}\ell'''(X_i;\theta^*).$$

Later in the semester we will formally show that $\big(\hat{\theta} - \theta_0\big) \xrightarrow{p} 0$. If the third derivative is bounded by $M(X)$, which has finite expectation, the strong law of large numbers implies that $\frac{1}{2n}\sum_i \ell'''(X_i;\theta^*)$ is bounded, and because $\big(\hat{\theta} - \theta_0\big) \xrightarrow{p} 0$, the entire remainder term $R_n \xrightarrow{p} 0$.
Finally, this combined work shows that the denominator in the expression for $\sqrt{n}\big(\hat{\theta} - \theta_0\big)$ is:

$$Y_n = -\frac{1}{n}\ell''(\theta_0) - \frac{1}{2n}\big(\hat{\theta} - \theta_0\big)\ell'''(\theta^*) \xrightarrow{p} \mathcal{I}(\theta_0),$$

and therefore:

$$\sqrt{n}\big(\hat{\theta} - \theta_0\big) = W_n/Y_n \xrightarrow{p} W/\mathcal{I}(\theta_0) \stackrel{d}{=} N(0, 1/\sqrt{\mathcal{I}(\theta_0)}),$$

or by multiplying by $\sqrt{\mathcal{I}(\theta_0)}$, we get the theorem statement:

$$\sqrt{n\mathcal{I}(\theta_0)}\big(\hat{\theta} - \theta_0\big) \xrightarrow{d} Z,$$

where $Z \sim N(0, 1)$. □

- How do we interpret / use the result above?
- *For large iid samples:* the MLE $\hat{\theta}$ is approximately normally distributed, centered at the *true* value θ_0, with shrinking variance:
$$\mathrm{Var}(\hat{\theta}) = \frac{1}{n\mathcal{I}(\theta_0)}.$$
- *Note* that there is sometimes some confusion regarding whether or not $\mathcal{I}$ is the information from *a single* observation, or *all observations*.
- Because the data are IID, it's really just a difference in scaling constants.
- In our definition and proof, we let $\mathcal{I}(\theta_0)$ be the information from a single observation:
$$\mathcal{I}(\theta_0) = E\Big[\Big(\frac{\partial}{\partial\theta}\log f(X_i;\theta)\Big)^2\Big].$$
- We could have instead considered the likelihood / information using all of the data points:
$$\mathcal{I}_n(\theta_0) = E\Big[\Big(\frac{\partial}{\partial\theta}\log f_{X_1:X_n}(X_1,\ldots,X_n;\theta)\Big)^2\Big].$$
- In the later case, the theorem would have stated:
$$\sqrt{\mathcal{I}_n}\big(\hat{\theta} - \theta_0\big) \xrightarrow{d} Z \sim N(0, 1),$$
or that
$$\mathrm{Var}(\hat{\theta}) = \frac{1}{\mathcal{I}_n(\theta_0)}$$
- In the IID case, these are the same, as
$$\begin{aligned}\mathcal{I}_n(\theta) &= -E\Big[\frac{\partial^2}{\partial\theta^2}\sum_i \log f(X_i;\theta)\Big] \\ &= -\sum_i E\Big[\frac{\partial^2}{\partial\theta^2}\log f(X_i;\theta)\Big] = n\mathcal{I}(\theta).\end{aligned}$$
- Our derivation treated θ as univariate. The more common multivariate case has a similar result:

Multivariate parameters
If $\theta \in \mathbb{R}^d$ is multivariate, than using the matrix definition of $\mathcal{I}(\theta)$, then:

$$\sqrt{n}\big(\hat{\theta} - \theta_0\big) \xrightarrow{d} Z_d \sim N_d\big(0_d, \mathcal{I}(\theta_0)^{-1}\big),$$

where $\hat{\theta}$ is the MLE, θ_0 is the true parameter value, and $N_d(\mu, \Sigma)$ is a d-dimensional multivariate normal distribution with mean vector μ and covariance matrix Σ.

4 Bias, MSE, and Consistency

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5 Sufficiency

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6 Efficiency

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